NRSur7dq4v2 PRECUSSING BBH WAVEFORM DATASET

Technical Data Sheet, Frame Dynamics, Sampling Methodology & Quality Benchmarks

Total Samples: $N = 102{,}400$
File Size: $\approx 15.9\,\text{GB}$ (HDF5)
Units: $(G = c = M = 1)$

1. Scientific Scope

Relativistic spin precession ($\vec{L} \cdot \vec{S}_{1,2}, \vec{S}_1 \cdot \vec{S}_2$) induces severe amplitude and phase modulations in inertial waveforms, causing spectral leakage and phase errors in neural networks. This dataset provides benchmark precessing waveforms to train deep generative models (e.g., `Aesur` autoencoders).

2. Architecture

HDF5 Path	Tensor Shape	Dtype	Description & Role in Physics / ML Pipeline
`/Waveforms`	$(102{,}400, 2048)$	`complex128`	Inertial frame $(2, 2)$ mode.
`/Waveforms_coorb`	$(102{,}400, 2048)$	`complex128`	Co-orbital $(2, 2)$ mode.
`/Parameters`	$(102{,}400, 7)$	`float32`	$q, \chi_{1x}, \chi_{1y}, \chi_{1z}, \chi_{2x}, \chi_{2y}, \chi_{2z}$ at t_{ref}
`/Dynamics/q_copr`	$(102{,}400, 4, 2048)$	`float32`	Unit quaternions.
`/Dynamics/chiA`	$(102{,}400, 2048, 3)$	`float32`	Primary spin trajectory $\vec{\chi}_A(t)$ in the coprecessing frame.
`/Dynamics/chiB`	$(102{,}400, 2048, 3)$	`float32`	Secondary spin trajectory $\vec{\chi}_B(t)$ in the coprecessing frame.
`/Dynamics/orbphase`	$(102{,}400, 2048)$	`float32`	Accumulated orbital phase $\phi_{\text{orb}}(t)$ in the coprecessing frame.

Temporal Grid: $t \in [-3996.0, +98.0]\,M$, step $\Delta t = 2.0\,M$, length $L = 2048$. The merger peak ($t = 0.0\,M$) is aligned at index 1998.

3. Surrogate Evaluator Settings & Parallel Generation Pipeline

Waveforms are generated using `gwsurrogate` evaluating the NR surrogate `NRSur7dq4v2`:

- `sur(q=q,chiA0=chiA0,chiB0=chiB0,f_low=0.0,times=times,ellMax=2,precessing_opts={'return_dynamics':True})`.
- The 102,400 configurations are partitioned across 4 concurrent worker processes. Workers process batches in isolated HDF5 part-files, followed by merge into final file.

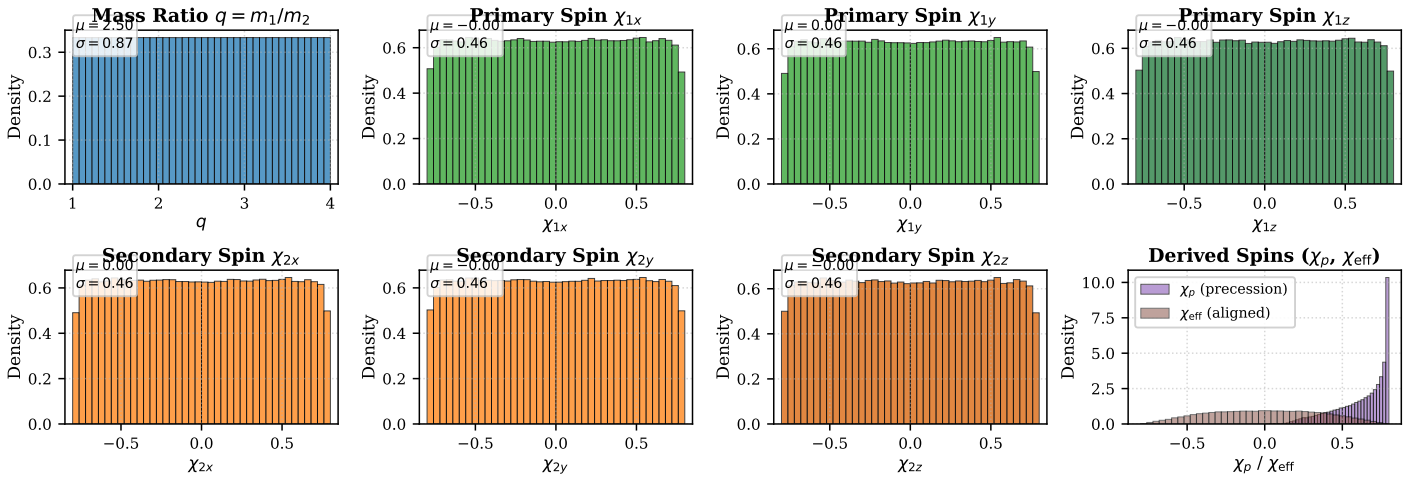
4. 7D Sampling Methodology: Scrambled Sobol + Optimal Transport Ball (OT-Ball)

Mass Ratio $q \in [1.0, 4.0]$**:** Sampled via a 1D Scrambled Sobol quasi-random sequence ($S_n \subset [0, 1)$, seed 42) mapped linearly $q = 1 + 3S_n$. This yields uniform 1D stratification ($\eta_H = 1.0000$) with minimal gap discrepancy.

Optimal Transport Ball (OT-Ball):

1. Seed $u_i \sim 2\,\text{Beta}(0.5, 0.5) - 1$, pushed to 3D sphere via squircular mapping.
2. Rank-order points in each Cartesian axis and map to equispaced target grid $T_j \in [-1.04\,a_{\max}, 1.04\,a_{\max}]$ with sub-grid dither.
3. Any point exceeding $|\vec{\chi}| > 0.8$ is projected radially back to $|\vec{\chi}| = 0.8(1 - 10^{-5})$.

5. Parameter Distributions (Histograms Across All 7 Coordinates)



6. Global Empirical Parameter Statistics ($N = 102{,}400$)

Summary statistics computed across the full dataset for neural network preprocessing and loss calibration:

Parameter	Col	Min	Max	Mean μ	Std σ	Flatness η_H	Physical Interpretation & Role in Network Training
$q = m_1/m_2$	0	1.0000	4.0000	2.5000	0.8660	1.0000	Mass ratio
χ_{1x}, χ_{1y}	1, 2	−0.7999	+0.7999	0.0000	0.4566	0.9993	Primary in-plane spin components.
χ_{1z}	3	−0.7999	+0.7999	0.0000	0.4566	0.9994	Primary aligned spin component.
χ_{2x}, χ_{2y}	4, 5	−0.7999	+0.7999	0.0000	0.4566	0.9993	Secondary in-plane spin components.
χ_{2z}	6	−0.7998	+0.7999	−0.0001	0.4567	0.9993	Secondary aligned spin component.
$\lvert\vec{\chi}_1\rvert, \lvert\vec{\chi}_2\rvert$	—	0.0241	0.8000	0.7885	0.0612	—	Dimensionless Kerr spin magnitudes ($a \lesssim 0.8$).
χ_{eff}	—	−0.7983	+0.7973	−0.0000	0.3512	0.9631	Mass-weighted aligned spin $(q\chi_{1z} + \chi_{2z})/(1 + q)$.
χ_p	—	0.0379	0.7999	0.6317	0.1631	0.8353	Dominant precession parameter.